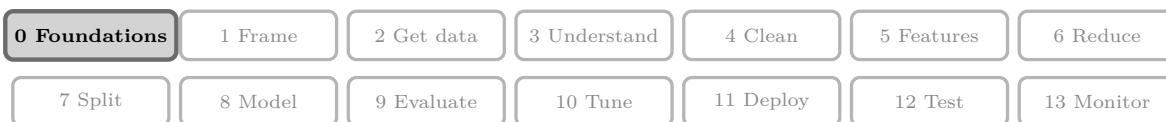


Matrix Multiplication as Composition

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Where this fits: Pipeline map step 0 (Foundations). See the [Course map](#).



- **Builds on:** Dot product ([Note 362](#)); Linear transformations and matrices ([Note 500](#)).
- **Leads to:** Jacobian and matrix gradients ([Note 602](#)); Neural networks ([Note 1002](#)); Forward propagation ([Note 1010](#)).

1. Overview

Key point: Multiplying two matrices gives the matrix of “apply one transformation, then the other”; we read the product from right to left.

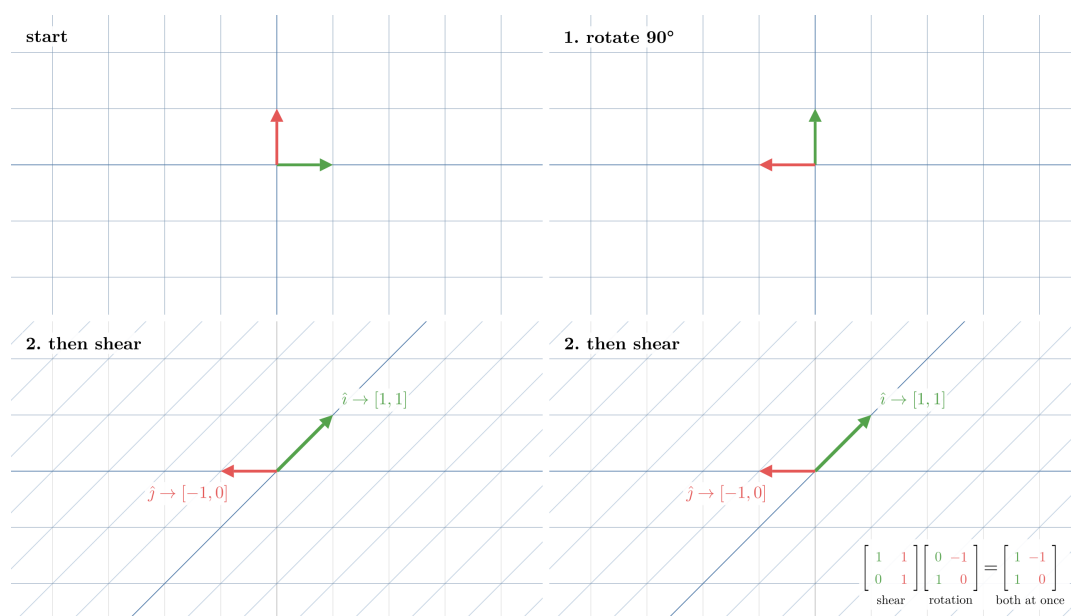


Figure 1: Rotating the plane by 90° and then shearing it is one single transformation; its matrix is the product of the two matrices

Figure 1 applies two transformations one after the other: a rotation by 90° , then a shear. From start to finish, the plane has undergone a third linear transformation, with a matrix of its own. That matrix is the product of the shear matrix and the rotation matrix.

This Note builds on the [linear transformations and matrices Note](#): the columns of a matrix are where $\hat{i}$ and $\hat{j}$ land, and $A\mathbf{x}$ is the linear combination of those columns.

2. Composition: one transformation after another

Key point: Applying one linear transformation and then another is again a linear transformation, called their composition; following $\hat{i}$ and $\hat{j}$ gives its matrix.

We often want to apply one transformation and then another. The overall effect is again a linear transformation (grid lines are still parallel and evenly spaced, and the origin has not moved). It is called the **composition** of the two transformations.

Like any linear transformation, it has a matrix, found by following $\hat{i}$ and $\hat{j}$ to their final places. In Figure 1:

- **Rotation by 90°** has the matrix R with columns $[0, 1]$ and $[-1, 0]$: it sends $\hat{i}$ to $[0, 1]$ and $\hat{j}$ to $[-1, 0]$.
- **The shear** has the matrix S with columns $[1, 0]$ and $[1, 1]$: it keeps $[1, 0]$ fixed and sends $[0, 1]$ to $[1, 1]$.
- **After both**, $\hat{i}$ has ended at $[1, 1]$ and $\hat{j}$ at $[-1, 0]$.

So the composition has the matrix with those two columns:

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}$$

This one matrix does in a single step what the rotation and the shear do in two.

3. Why the product is written right to left

Key point: In $SR\mathbf{x}$, the matrix next to the vector acts first: R is applied first, then S .

Applying the rotation and then the shear to a vector $\mathbf{x}$ the long way means two multiplications: first $R\mathbf{x}$, then S times that result, $S(R\mathbf{x})$. The composition matrix must give the same answer for every $\mathbf{x}$, since it describes the same overall movement. So it is natural to call it the **product** SR :

$$S(R\mathbf{x}) = (SR)\mathbf{x}$$

Note the order. The transformation on the right (R) happens first and the one on the left (S) second. This comes from function notation: in $f(g(x))$, g acts first. A matrix product is read from right to left.

4. Computing a product column by column

Key point: The first column of M_2M_1 is M_2 times the first column of M_1 ; the second column is M_2 times the second column of M_1 .

We can find the product without watching any animation, from the numbers alone. Take

$$M_1 = \begin{bmatrix} 1 & -2 \\ 1 & 0 \end{bmatrix}, \quad M_2 = \begin{bmatrix} 0 & 2 \\ 1 & 0 \end{bmatrix}$$

and apply M_1 first, then M_2 .

1. **In words:** after M_1 , $\hat{i}$ sits at the first column of M_1 . Applying M_2 to that column gives where $\hat{i}$ finally lands: the first column of the product. Do the same with the second column for $\hat{j}$.
2. **Formula:**

$$M_2 M_1 = \left[M_2 \cdot (\text{column 1 of } M_1) \mid M_2 \cdot (\text{column 2 of } M_1) \right]$$

3. **Example:**

- $\hat{i}$ first lands on $[1, 1]$. Then $M_2[1, 1] = 1 \cdot [0, 1] + 1 \cdot [2, 0] = [2, 1]$.
- $\hat{j}$ first lands on $[-2, 0]$. Then $M_2[-2, 0] = -2 \cdot [0, 1] + 0 \cdot [2, 0] = [0, -2]$.

$$M_2 M_1 = \begin{bmatrix} 0 & 2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 1 & -2 \end{bmatrix}$$

The same reasoning with letters gives the general rule. Take

$$M_1 = \begin{bmatrix} e & f \\ g & h \end{bmatrix}, \quad M_2 = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

The first column of the product is $e[a, c] + g[b, d]$ and the second is $f[a, c] + h[b, d]$:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} ae + bg & af + bh \\ ce + dg & cf + dh \end{bmatrix}$$

Extra: The same formula read entry by entry is the school rule “row times column”. The entry in row i , column j of AB is the dot product of row i of A with column j of B . In the example, row 1 of M_2 is $[0, 2]$ and column 1 of M_1 is $[1, 1]$, and $0 \times 1 + 2 \times 1 = 2$, the top-left entry. This is why the [dot product and cosine similarity Note](#) calls the dot product the building block of matrix multiplication.

Python: @ multiplies two matrices.

```
import numpy as np

M1 = np.array([[1, -2], [1, 0]])
M2 = np.array([[0, 2], [1, 0]])
M2 @ M1           # [[2, 0], [1, -2]]
x = np.array([1, 1])
M2 @ (M1 @ x)     # [2, -1]: two steps
(M2 @ M1) @ x     # [2, -1]: one step
```

5. Order matters

Key point: In general $AB \neq BA$: shearing then rotating is a different transformation from rotating then shearing.

Does the order of the two matrices matter? Thinking in transformations, we can answer by picturing them (Figure 2).

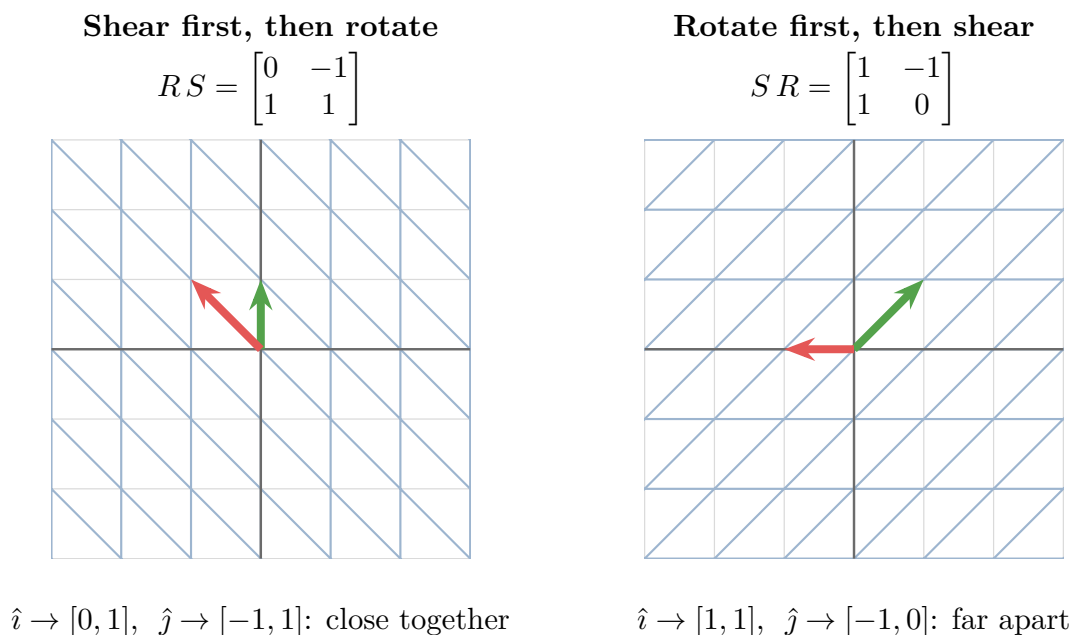


Figure 2: The same two transformations in the two possible orders

- **Shear first, then rotate** (RS): $\hat{i}$ ends at $[0, 1]$ and $\hat{j}$ at $[-1, 1]$. The two vectors point close together.
- **Rotate first, then shear** (SR): $\hat{i}$ ends at $[1, 1]$ and $\hat{j}$ at $[-1, 0]$. They point far apart.

The overall effects differ, so $RS \neq SR$:

$$RS = \begin{bmatrix} 0 & -1 \\ 1 & 1 \end{bmatrix} \neq \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix} = SR$$

Matrix multiplication is **not commutative**, unlike the multiplication of numbers. In code this means $A @ B$ and $B @ A$ are, in general, different matrices.

6. Grouping does not matter

Key point: $(AB)C = A(BC)$, because both mean “apply C , then B , then A ”.

For three matrices A , B and C , we can first compute AB and multiply the result by C , or first compute BC and put A in front. Both give the same matrix:

$$(AB)C = A(BC)$$

This property is called **associativity**. Proving it by expanding all the entries is long and tells us nothing. Seen as transformations it is immediate: both sides mean “apply C , then B , then A ”, the same three steps in the same order. Only the bookkeeping differs.

Python: A quick check with the three matrices of this Note.

```
R = np.array([[0, -1], [1, 0]])
S = np.array([[1, 1], [0, 1]])
np.array_equal((S @ R) @ M1, S @ (R @ M1))    # True
np.array_equal(S @ R, R @ S)                  # False
```

7. Where ML multiplies matrices

Key point: A product needs matching inner sizes; stacked linear layers collapse into one matrix, which is why neural networks need activation functions between them.

7.1 The shape rule

Key point: An $m \times n$ matrix times an $n \times p$ matrix gives an $m \times p$ matrix; the inner sizes must match.

So far all matrices were 2×2 . In ML the matrices are rectangular, and the column-by-column view explains the shape rule. In AB , each column of B is a vector that A transforms, so each column of B must have as many entries as A has columns.

1. **In words:** the number of columns of the left matrix must equal the number of rows of the right matrix; the result takes the rows of the left and the columns of the right.
2. **Formula:**

$$(m \times n)(n \times p) = (m \times p)$$

3. **Example:** the data matrix X with 3 points and 2 features times A^T in the [linear transformations and matrices Note](#) (section 7.1) is $(3 \times 2)(2 \times 2) = 3 \times 2$: three transformed points. PCA's XW^T with 40 points, 3 features and 2 components is $(40 \times 3)(3 \times 2) = 40 \times 2$ (see the [PCA step by step Note](#), section 5.1).

The dot product $a^T b$, $(1 \times n)(n \times 1) = 1 \times 1$, is the smallest case of this rule.

7.2 Linear layers collapse into one

Key point: Two layers without an activation, $W_2(W_1 \mathbf{x})$, are the single layer $(W_2 W_1) \mathbf{x}$: stacking them adds no power.

A neural network applies layer after layer. Suppose two layers had no activation function between them, only the matrices W_1 and then W_2 . By Section 3, the two steps are one transformation:

$$W_2(W_1\mathbf{x}) = (W_2W_1)\mathbf{x}$$

With the numbers of Section 4, $W_1 = M_1$ and $W_2 = M_2$: the input $[1, 1]$ goes to $M_1[1, 1] = [-1, 1]$ and then to $M_2[-1, 1] = [2, -1]$. The single matrix M_2M_1 sends $[1, 1]$ straight to $[2, -1]$. However many linear layers we stack, the result is still one matrix, one linear transformation.

This is why every layer of a network ends with a non-linear activation such as the [sigmoid](#). The activation bends the space between layers, so the composition can no longer be squeezed into a single matrix. The classic example is XOR: no linear model can fit it, while a network with one hidden layer and a non-linear activation can (Goodfellow et al. §6.1).

Extra: The same collapse shows in linear regression with engineered features. If every new feature is a linear combination of the old ones, the model can learn nothing new: the combined effect is still a linear combination of the original columns. Polynomial features (see the [polynomial regression Note](#)) help precisely because squaring is not linear.

8. Summary

Idea	What it says	Example
Composition	one transformation after another	rotate, then shear
Product BA	matrix of “apply A , then B ”	$SR = \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}$
Column by column	column j of BA is B times column j of A	$M_2[1, 1] = [2, 1]$
Not commutative	$AB \neq BA$ in general	$RS \neq SR$
Associative	$(AB)C = A(BC)$	same three steps, same order
Shape rule	$(m \times n)(n \times p) = m \times p$	$(40 \times 3)(3 \times 2) = 40 \times 2$

- A matrix product is a composition of transformations, read from right to left.
- Follow $\hat{i}$ and $\hat{j}$ through both steps to get the columns of the product.
- Order matters; grouping does not.
- Stacked linear layers collapse into one matrix; activation functions prevent that.

Sources

- Goodfellow, I., Bengio, Y. and Courville, A. (2016). *Deep Learning*. MIT Press. Section 6.1, learning XOR.

9. Key terms

Term	Meaning
Composition	Applying one transformation and then another, seen as one overall transformation
Matrix product	The matrix BA of the composition “apply A , then B ”
Not commutative	The order of the factors matters: $AB \neq BA$ in general
Associativity	$(AB)C = A(BC)$: the grouping of a product does not matter
Shape rule	$(m \times n)(n \times p) = m \times p$; the inner sizes must match